

BANKING SYSTEM DOCUMENTATION (JAVA + SQL)

SOLUTIONS BANKING

Team:

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Banking Management System using Java and MySQL.

Introduction

The Banking Management System is a software application developed to help banks manage customer accounts and transactions digitally.

The system allows users to:

- Create customer accounts
- Deposit money
- Withdraw money
- Transfer funds
- Check balances
- Store banking information securely in a database

The project uses:

- NetBeans to code for JAVA (System)
- XAMPP to code for MySQL (database management)
- Android studio to code for Kotlin (Application)
- JDBC for connecting Java to MySQL

Objectives

The main objectives of the system are:

- To automate banking operations
- To provide fast banking services
- To improve transaction accuracy
- To securely store customer records
- To reduce manual paperwork

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System Features

Main Features

Customer Management

- **Add new customer**
- **View customer details**
- **Update customer information**

Account Management

- Create bank accounts
- View account information
- Check balances

Database Management

- Store records permanently
- Maintain relationships between tables
- Ensure data integrity

Technologies Used

Java – Backend programming

MySQL – DATABASE

JDBC – Database connection

NetBeans – IDE

XAMPP – Database server

System Architecture

The system follows a simple architecture:

1. User enters information
2. Java application processes data
3. JDBC connects to MySQL database
4. Database stores or retrieves information
5. Results displayed to user

Database Design

- **Create database**
- **Create table for:**
 1. **Customer management**
 2. **Account management**
 3. **Transactions**
- **Insert into for:**
 1. **Customer management**
 2. **Account management**
 3. **Transactions**

Java Design

- **Customer register, view & update**
- **Account creating & managing different account types**
- **Transaction handling deposits, withdrawals & transfer between accounts**
- **Balance enquiry allowing customers or staff to check account balances**

Conclusion

The Banking Management System successfully demonstrates how Java and MySQL can be integrated to automate banking operations. The system improves efficiency, accuracy, and security while reducing manual work. It provides a strong foundation for developing more advanced banking applications in the future